Where to look for life and technology: An audited decision analysis of confirmed exoplanets

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Abstract

Which confirmed exoplanets should be prioritized when both environmental plausibility and observational access matter? Individual probabilities of life or civilization are not identified by present data, so a numerical "probability of life" ranking would obscure the dominant uncertainty. We instead construct a reproducible, conditional decision audit. A frozen NASA Exoplanet Archive query contains 6,451 planet records; explicit provenance, status, size and flux checks leave 190 planets for an Earth-adjacent atmospheric screen. Fifteen sourced parameter overrides correct distances, reconcile stellar measurements and preserve dynamical mass information. We propagate 8,192 physical realizations per planet over 128 orbital phases through 648 environmental and instrument scenarios. Eligibility and recoverability are evaluated on the same latent states, and a global minimax calculation examines equal-time portfolios. Proxima Cen b ranks first in 92.4% of scenarios accessible to a southern 39 m telescope. GJ 887 d supplies an alternative when larger temperate worlds are admitted. A corrected distance nearly doubles the nominal angular separation of Gl 725 B c, but its declination excludes it from the southern telescope. A separate injection experiment uses ten published photochemical transmission spectra. At 20 ppm white plus 5 ppm correlated noise, recovery among six inhabited models ranges from 2.99% to 89.36%, despite complete atmospheric-feature recovery in the simulated trials. Omitting airless alternatives yields a 99.53% false-positive rate for a flat spectrum; admitting them reduces that rate to 0.035%. A radio experiment covering 4,792 confirmed-planet hosts shows strong dependence on assumed transmitter power, duty cycle and persistent beaming. The deliverable is a set of auditable priorities and discriminating follow-up questions, not a claim that any world is probably inhabited or that a modeled molecular feature establishes life.

Keywords: exoplanets; biosignatures; technosignatures; target selection; uncertainty; injection and recovery

1 The scientific question and the contribution

The attractive target is often described as a planet that is both likely to host life and easy to observe. The two parts of that description have very different evidential foundations. Distance, orbital period, apparent brightness, and sometimes mass and radius are measured. Atmospheric survival, surface water, the emergence of life, biological productivity, and technological activity are usually unknown. An observing program must act despite this asymmetry. It should preserve the uncertainty rather than replace it with a precise-looking score.

This study asks a narrower, operational question: *which confirmed planets remain useful observing targets across explicitly different environmental preferences and observational assumptions, and which apparently favorable cases fail when the source data or the null hypotheses are audited?* The restriction to confirmed planets refers to a dated archive and a stated status rule. It does not establish that every retained signal is indisputable, nor that other undiscovered planets in the same systems are absent.

The distinction between habitability, detectable biology, and evidence for biology is well established (Catling et al., 2018; Walker et al., 2018). Earth-history arguments can inform particular models of abiogenesis, but depend on conditioning and model assumptions and do not calibrate the relative probability of life on an arbitrary M-dwarf planet and a water-rich super-Earth (Spiegel & Turner, 2012; Kipping, 2020). The same difficulty is larger for civilization: neither technological emergence, survival, transmitter design, nor the fraction of time a signal crosses Earth is measured for another civilization. We consequently avoid assigning empirical probabilities to these unknown steps.

1.1 What is already known, and what is new here

Habitable-zone catalogs, orbital completeness calculations, and simulated atmospheric spectra are extensive areas of prior work. Single-visit completeness combines geometry and photometric access (Brown, 2005); orbital Monte Carlo has long propagated phase, inclination and brightness (Greco & Burrows, 2015). LIFE X already applies such reasoning to the confirmed-planet catalog and compares future thermal and reflected-light access (Carrión-González et al., 2023). Relative biosignature yields, population simulations, and information-based survey design are developed by Tuchow & Wright (2020), Bixel & Apai (2021), and Fields et al. (2023). The recent rocky habitable-zone catalog of Bohl et al. (2026) is a particularly close target-list comparator. Photochemical simulations already include inhabited atmospheres, abiotic false positives, and different observing modes (Currie & Meadows, 2025; Eager-Nash et al., 2024a).

Our contribution is the *new numerical audit of their conjunction*: a frozen 2026 confirmed catalog with traceable corrections; the measured size of the error from multiplying separately averaged eligibility and observability; scenario-dependent ranks and an exact portfolio optimization; a finite-family spectral test that explicitly challenges missing alternatives; and a radio calculation that separates observing cost from occurrence assumptions. We do not claim the first habitable-zone ranking, the first inhabited-versus-abiotic spectrum, or the first Bayesian interpretation of a SETI null. Nor do we assert that an individual candidate was overlooked by all prior literature. A planet can be underweighted by a particular catalog query without having been overlooked by its discovery authors.

Three simulation modules are deliberately kept distinct. The catalog module estimates recovery of a *stipulated* spectral feature. The atmospheric module measures discrimination within published model families at achieved noise levels. The radio module measures recovery of specified narrowband transmitter populations. Their numerical outputs are not interchangeable, and multiplying them into an overall life probability would require additional, unsupported population assumptions.

1.2 Time horizon and scope

We use 2026–2036 as the near-future planning interval. Radio facilities and JWST already support relevant observations. A 39 m ground-based scenario is conditional on the relevant instrumentation becoming available: ESO lists planned ELT telescope first light in 2029 and scientific first light in 2030, while ANDES is a later instrument (ESO, 2025; Pallé et al., 2025). The high-contrast ANDES mode is specified in approximately 0.95–1.8 μm, even though the full spectrograph covers a wider range (ANDES Consortium, 2026). We therefore use 1.27 μm for the ground-based comparison and do not assign the visible oxygen A band to the baseline high-contrast mode.

A 6 m Habitable Worlds Observatory (HWO) comparison is included only to reveal a geometric change with aperture and wavelength. HWO remains an architecture-development program; it is not assumed operational within our near-future interval (NASA, 2026). LIFE offers a complementary longer-horizon thermal route, but we do not implement a new interferometric noise model. Broadband thermal detection forecasts must not be substituted for atmospheric retrieval requirements (Carrión-González et al., 2023; Konrad et al., 2022).

2 Estimands: what can and cannot be ranked

Let x_j be the available data for planet j, z_j its latent physical state, L_j the presence of life, and a an observing design. A discovery-oriented quantity would be

$$U_j(a) = \int p(z_j \mid x_j)\, p(L_j = 1 \mid z_j)\, P(\text{validated life evidence} \mid L_j, z_j, a)\, \mathrm{d}z_j. \tag{1}$$

Neither the second factor nor the complete third factor is known. In particular, an atmospheric line can be recovered when both biotic and abiotic models fit it. Even perfect precision cannot distinguish hypotheses that predict identical data distributions. The archive alone therefore does not identify a most-probably-inhabited planet. Different admissible biological assignments can reverse its ordering without changing any measured catalog value.

We replace Eq. (1) with a declared *conditional decision weight*,

$$\mathcal{J}_{jm} = \mathbb{E}_{z_j|x_j,m}[w_m(z_j) D_m(z_j, a)]\,, \qquad 0 \le w_m, D_m \le 1, \tag{2}$$

where m indexes the scenario, w_m encodes environmental eligibility and preferences, and D_m is the power to recover the stipulated feature under the observation surrogate. A value of $\mathcal{J} = 0.1$ is *not* a 10% life probability. It is an expectation within an explicitly chosen model. We retain the underlying eligibility and recovery terms so that a reader can substitute different preferences.

2.1 Why separate averages can be misleading

The shortcut $\widetilde{\mathcal{J}}_{jm} = \mathbb{E}[w_m]\mathbb{E}[D_m]$ discards shared states. The exact identity

$$\mathcal{J}_{jm} - \widetilde{\mathcal{J}}_{jm} = \mathrm{Cov}(w_m, D_m) \tag{3}$$

is elementary, but its numerical size for catalog target selection deserves measurement. A nearly face-on radial-velocity solution can be geometrically favorable while implying a much larger true mass. A larger inferred radius brightens the planet while a mass gate excludes the same realization. An orbit or luminosity realization can improve reflected flux while moving outside a temperature-related eligibility interval. Coupling all factors to the same draw retains these effects. Independent shuffling is a useful negative control, not a physical alternative universe.

The uncertainty model used below is a simulation distribution assembled from published marginal uncertainties and declared completions. It is not a reconstruction of every original joint likelihood. Accordingly, Eq. (2) should not be read as a fully hierarchical posterior estimate. The notation emphasizes conditioning and shared variables; the limitations of the approximation remain material.

2.2 Robustness is a property of a stated scenario set

For each scenario we normalize by its best target and record ranks. The fraction of scenarios in which a planet ranks in the first five is a *scenario frequency*; equal weighting of the grid is a reporting convention. It is not a posterior probability, and it would change if the scenario dictionary were expanded or duplicated.

For an equal-time portfolio A of three different planets, each receiving 100 hours, define

$$R(A) = \max_m \left[1 - \frac{\sum_{j \in A} \mathcal{J}_{jm}}{\sum_{j \in A^\star_m} \mathcal{J}_{jm}} \right], \tag{4}$$

where $A^\star_m$ contains the three best planets in scenario m. A mixed-integer linear program minimizes $R(A)$ over all candidates with a nonzero score. The sum is an additive conditional utility; it does not claim

that discovery events or biospheres are independent. Optimizing the chance of at least one life detection would require those additional probabilities and correlations. The oracle is hypothetical knowledge of the scenario, not a feasible adaptive observation schedule.

3 A frozen catalog and an explicit evidence audit

We queried the NASA Exoplanet Archive Planetary Systems Composite Parameters table on 17 October 2026 at 17:17 UTC, retrieving 113 columns for 6,451 planet rows and 4,847 host labels (NASA Exoplanet Archive, 2026). The composite table increases completeness by combining literature values; it is not guaranteed to represent one internally consistent stellar/planetary solution. The archive query, unmodified CSV, source references, retrieval timestamp and SHA-256 checksum accompany this manuscript.

The initial status rule requires the archive controversy flag to be zero, leaving 6,393 planet rows. The atmospheric screen then requires either a measured transit radius $R_p \leq 3R_\oplus$, or, where no measured radius is available, a measured mass or minimum mass $\leq 10M_\oplus$. Archive radii marked as calculated, mass–radius predictions, and quoted upper limits are not treated as direct measurements. A radius is admitted as measured in this screen only when a transit and a non-calculated, non-limit radius are present. This intentionally excludes unsupported radius claims for nearby nontransiting radial-velocity planets.

We require positive distance and semimajor axis, a host temperature between 2,500 and 7,200 K, and a luminosity estimate. Missing semimajor axes are completed from orbital period and stellar mass; missing luminosities from radius and temperature. We recompute incident flux rather than accepting an isolated insolation cell:

$$S_a = \frac{L_*/L_\odot}{(a/\mathrm{AU})^2}, \qquad \overline{S} = \frac{S_a}{\sqrt{1-e^2}}, \tag{5}$$

where the second expression is the orbital time average for a Keplerian orbit. The broad screen is $0.1 \leq \overline{S} \leq 3$ in Earth-flux units. It is a computational inclusion rule, not a claim that the entire interval permits a habitable surface. Nominal cuts can miss uncertainty tails; we do not infer population occurrence from the selected set.

After the numerical updates below, the mechanical screen contains 191 planets. We retain a traceable exception for GJ 682 b: although unflagged in the frozen archive, a primary radial-velocity reanalysis failed to recover the proposed signals and conservatively treated the system as having no planets (Tuomi et al., 2019). We exclude it from the curated atmospheric core while preserving its catalog row and the reason. This leaves **190 planets in 167 host systems**, including 66 planets within 30 pc and 133 with measured transit radii. Nineteen have both measured mass and radius; 43 fall nominally in our conservative habitable-zone interval and 88 in the empirical interval. These counts differ from narrower rocky catalogs because the scope, date, parameter reconciliation and composition gates differ.

3.1 Corrections that change decisions

Fifteen overrides are applied to a working copy, with each original value retained. They include three distance entries; coherent stellar luminosity, radius and temperature updates for three hosts; a shared inclination for GJ 1061; and eccentricity plus pericenter updates for Ross 128 b (Liebing et al., 2024). They are sourced corrections or alternative stellar measurements, not adjustments made to obtain a preferred ranking. Table 1 summarizes the most consequential cases.

For GJ 273, the original large luminosity uncertainty is present in the discovery literature; it is not an archive transcription error. We use an independent SED/interferometric determination (Rabus et al., 2019). Luminosity and radius are rescaled to the adopted parallax as $L \propto d^2$ and $R \propto d$, retaining the

Table 1: Selected audit findings. Angular separation is the nominal a/d scale, not a guarantee of access at an observation epoch. All calculations preserve measured apparent brightness.

Object	Audited input or identity	Consequence and provenance
GJ 273	$d : 5.9215 \rightarrow 3.7861$ pc; updated L_*, R_*, T_*	$a/d : 15.4 \rightarrow 24.1$ mas for b. A more precise stellar solution gives $\overline{S} \simeq 1.23$, warmer than the adopted conservative interval. Gaia EDR3 (Gaia Collaboration, 2020); Rabus et al. (2019).
Gl 725 B	$d : 6.8424 \rightarrow 3.5230$ pc	$a/d : 20.3 \rightarrow 39.5$ mas for c; radio power threshold improves by 3.77×. Northern declination excludes this target from the southern 39 m scenario. Cristofari et al. (2022); Ould-Elhkim et al. (2026).
Wolf 1061	$L_* \simeq 0.01109 L_\odot$ with much tighter uncertainty	Removes a broad artificial tail of conservative-zone eligibility for c; the planet still enters warm-edge empirical scenarios. Interferometry and SED: Kane et al. (2017).
GJ 3323	$L_* = 0.003606 L_\odot$ from measured bolometric flux and distance	c moves to $S_a \simeq 0.226$; b moves beyond the broad upper flux cut. Mann et al. (2015).
GJ 1061	Shared $i = 77^{+9}_{-13}$ deg in the adopted dynamical solution	c and d have model-dependent true masses; dividing those masses again by $\sin i$ would be incorrect. Dreizler et al. (2025).
HD 20794	Archive f ↔ paper d at $P = 647.6$ d	Match host and period. The archive's d label refers to a different, approximately 90-day signal. Nari et al. (2025).
GJ 414 A b	Nontransiting; catalog radius is a prediction	Preserve its radial-velocity mass information and do not infer a measured terrestrial radius. Dedrick et al. (2021).

published fractional uncertainty conservatively. The Wolf 1061 update follows the same procedure using Kane et al. (2017). For GJ 3323, the measured bolometric flux of Mann et al. (2015) is converted using $L = 4\pi d^2 F_{\rm bol}$, and radius follows from the Stefan–Boltzmann relation. These are coherent nominal solutions, although full correlated uncertainty matrices are unavailable.

Distance corrections have different observational consequences in different channels. Reflected-light photon counts use measured J magnitude, so they must not receive an additional inverse-square brightness boost when distance changes. The corrected angular separation affects access and stellar leakage. A radio power threshold is an inferred source luminosity for a fixed received flux, so its d^2 scaling does change directly.

The audit also identifies a strongly inconsistent insolation entry for TOI-178 f; recomputed luminosity and orbit remove it from the temperate screen. Source identity matters beyond isolated numbers: GJ 887 d is now a confirmed 50.77-day planet (Hartogh et al., 2026); GJ 3378 b has a revised 21.45-day orbit and lower minimum mass (Robertson et al., 2026); and the archive includes GJ 251 c despite "candidate" wording in its discovery paper (Beard et al., 2025). We state those distinctions rather than silently mixing epochs or labels.

4 Catalog simulation and observing surrogates

4.1 Physical states, uncertain inclinations, and missing parameters

The main run draws 8,192 physical states for each of 190 planets, with 128 equally spaced mean-anomaly nodes per state. The result is 1,556,480 physical draws and 199,229,440 orbital state evaluations, reused across the scenario grid. Phase nodes are deterministic integration points, not independent posterior samples.

Quoted upper and lower uncertainties define a two-sided Gaussian perturbation: a standard normal variate is multiplied by the corresponding signed-side scale, then physically invalid states are rejected. This preserves the reported asymmetric scales but is not a recovered publication posterior. Missing errors are replaced by explicit defaults: 3% in semimajor axis, 0.05 dex in luminosity, 75 K in temperature, 1% in distance, 20% in mass, and 8% in radius. Measured eccentricities are sampled on $[0, 0.85]$; otherwise a Rayleigh distribution of scale 0.05 is used, truncated by any valid upper limit. Fixed-zero eccentricities are therefore treated as uncertain rather than as measured circularity. A sensitivity test uses scales zero and 0.15.

For a nontransiting radial-velocity planet with unknown inclination, true mass and geometry share the same i:

$$M_p = \frac{m_0}{\sin i}, \qquad m_0 = M_p \sin i. \tag{6}$$

At fixed m_0, an isotropic orientation prior and a mass-density factor proportional to $M^{-\gamma}$ induce $p(i \mid m_0, \gamma) \propto \sin^\gamma i$. We use $\gamma = 0, 1, 2$ as a *local inclination-prior stress test*. Because the measured m_0 is also perturbed without reconstructing its full likelihood and reference prior, this is not a complete reanalysis under three mass-population posteriors.

To avoid unstable weights near a face-on orbit, states are drawn from the bounded mixture proposal

$$q(i) = \tfrac{1}{2} \sin i + \frac{1}{\pi}, \quad 0 \le i \le \frac{\pi}{2}, \qquad \omega_\gamma(i) \propto \frac{\sin^\gamma i}{q(i)}. \tag{7}$$

This mixes isotropic and uniform-inclination draws. Every reported scenario average uses its matching normalized importance weights; baseline observability uses $\gamma = 1$. The weights are bounded for the tested γ values. Analytic inclination moments independently check the implementation.

Measured transit inclinations are retained. When a transiting object lacks one, a uniform impact parameter between zero and one gives a circular transit-conditioned approximation. For GJ 1061 the adopted common inclination is drawn once per system realization, and catalog dynamical true masses are retained without a second deprojection. The missing mass–inclination covariance in that solution remains a limitation (Dreizler et al., 2025).

For an unmeasured radius we compare two compositions. The compact case uses the Zeng et al. (2016) relation with core mass fraction 1/3,

$$R_p / R_\oplus = (M_p / M_\oplus)^{1/3.7}, \tag{8}$$

with 0.05 scatter in natural log radius, through $10 M_\oplus$. The published fit is calibrated for 1–8 $M_\oplus$; its use below 1 and from 8 to 10 $M_\oplus$ is a stipulated extrapolation. A bounded continuation above $10 M_\oplus$ is needed only for marginal observability; such draws fail every mass eligibility gate. The second case uses a continuous, two-slope population surrogate inspired by Chen & Kipping (2017): $R = 1.008 M^{0.279}$ below $2.04 M_\oplus$ and a continuous $M^{0.589}$ branch above, with 0.15 log scatter. It is not the Forecaster posterior. For transiting planets with no measured mass, the compact inverse relation with 0.3 log scatter supplies an assumed high-density completion; genuine mass upper limits truncate this distribution rather than create a point mass at the limit. High density itself reduces scale height at fixed radius. No imputed radius is subsequently described as a measurement.

4.2 Environmental scenario grid

Each environment weight is a product of an eligibility indicator and a chosen host-type weight:

$$w_m = \mathbf{1}(\text{size gate})\, \mathbf{1}(\text{flux/climate gate})\, s_m(T_*). \tag{9}$$

This product defines a preference rule; it is not a factorization of empirically independent biological causes. In particular, age, activity, stellar type and atmospheric escape are not multiplied as independent survival probabilities.

The size gates are $(M_{\max}/M_\oplus, R_{\max}/R_\oplus) = (3, 1.6), (5, 2), (10, 3)$, named small, standard and extended. A measured radius uses its radius gate; otherwise a true-mass draw uses its mass gate. The extended case admits larger temperate worlds without assuming their surfaces resemble Earth.

Conservative and empirical flux intervals use the one-Earth-mass polynomial boundaries of Kopparapu et al. (2013, 2014). The conservative interval runs from maximum greenhouse to runaway greenhouse; the empirical interval from early Mars to recent Venus. Coefficients are included in the reproducibility package. Temperature is clamped to the calibrated 2,600–7,200 K range; TRAPPIST-1 lies slightly below the lower boundary. These polynomials are not a climate model for each composition or planet mass.

For each interval we compare an orbit-mean criterion using Eq. (5) with a condition requiring at least 90% of orbital time inside the flux interval. The latter is a climate-persistence stress test, not a claim that organisms die whenever instantaneous flux crosses a boundary. Seasonal heat storage, atmospheric circulation, ocean depth, tidal heating, spin state and greenhouse feedbacks require dedicated climate calculations.

Host weights are flat $(1, 1, 1)$, mildly penalizing $(0.3, 0.7, 1)$, or strongly penalizing $(0.03, 0.3, 1)$ for $T_* < 3300$ K, $3300 \le T_* < 4000$ K, and warmer hosts, respectively. These factors are deliberately transparent. They are not atmospheric-retention probabilities, and hard temperature bins should not be interpreted as physical discontinuities. Current cosmic-shoreline extrapolations provide a reason to stress stellar assumptions, not a calibrated individual-world prior (Berta-Thompson et al., 2026). Counting three size gates, two flux intervals, two climate rules, three stellar weights, two radius completions, three inclination cases, and three instruments gives 648 scenarios per planet.

4.3 Reflected-light geometry and feature recovery

We solve Kepler's equation at each phase. With star–planet distance r, phase angle α, and observer distance d,

$$\theta = \frac{r \sin\alpha}{d}, \qquad C = A_g \Phi_L(\alpha) \left(\frac{R_p}{r}\right)^2, \quad \Phi_L(\alpha) = \frac{\sin\alpha + (\pi - \alpha)\cos\alpha}{\pi}. \tag{10}$$

The physical separation r, not projected separation, enters illumination. The stellar-reflex versus planetary pericenter convention is reconciled for radial-velocity arguments of pericenter. Geometric albedo is uniform from 0.05 to 0.35. A Lambert sphere and a fixed feature depth do not reproduce wavelength-dependent clouds or surface reflection; more detailed alternatives already exist (Currie & Meadows, 2025; Houelle et al., 2026).

The observing surrogate uses a 39 m aperture at $\lambda = 1.27\,\mu$m, throughput 0.05, total exposure 100 hours, and an effective feature width of $0.04\,\mu$m. Stellar photon counts N_* follow the observed J magnitude with a local blackbody color correction. The J-band zero point is 1,594 Jy at 1.235 μm (2MASS, 2008). A stipulated feature removes 20% of planet continuum flux over that effective interval. This is a transparent equivalent-feature approximation; it does not include resolved molecular line lists, telluric masks, detector noise, Doppler overlap, sky emission or a full high-resolution cross-correlation pipeline. Consequently, it is not an ANDES exposure-time forecast or an oxygen detection prediction.

The contrast uncertainty and noncentral signal-to-noise ratio are

$$\sigma_C^2 = \frac{C + \ell(\theta)}{N_*} + c_{\rm sys}^2, \qquad \rho = \frac{0.2C}{\sigma_C}, \tag{11}$$

Table 2: Ground-based performance stress cases, all at 1.27 μm. The consortium raw-contrast benchmark motivates the reference leakage; the remaining values are assumptions, not delivered specifications.

Case	IWA / (λ/D)	IWA (mas)	Leakage at 20 mas	$c_{\rm sys}$
Ambitious	1	6.72	1.5×10^{-4}	3×10^{-10}
Reference	2	13.43	1.5×10^{-3}	10^{-9}
Difficult	3	20.15	1.5×10^{-3}	3×10^{-9}

where raw stellar leakage contributes photon noise and $c_{\rm sys}$ is a nonintegrating residual contrast floor. The adopted leakage decreases as θ^{-2} from its value at 20 mas, with a floor at one-hundredth of that value. We keep this extrapolation explicit rather than treating the consortium's single quoted raw-contrast benchmark as a validated full curve.

For a one-sided five-sigma threshold, excess feature power is $\max\{0, \Phi_N(\rho - 5) - \Phi_N(-5)\}$, multiplied by the angular-access indicator, where Φ_N is the standard-normal cumulative distribution function, distinct from the Lambert phase law Φ_L. This baseline-subtracted utility differs from literal detection power by at most 2.87×10^{-7}. Subtracting the noise-only baseline prevents distant, signal-free objects acquiring nominal utility merely through false alarms. Random-phase recovery averages over phase; best-phase recovery takes the maximum within each physical realization. Both assume that the full exposure can be coadded at the chosen phase over repeated orbital cycles. They do not describe a continuous 100-hour stare at a stationary orbital position. Real ephemerides, seasons, weather, telluric velocity separation and scheduling may substantially reduce feasibility.

A southern-site access cut uses latitude -24.6 deg and requires maximum altitude of at least 30 deg. It is a coarse visibility gate, not an airmass or calendar-time integration. All-sky scores are retained only as comparisons for an otherwise similar observing capability. For HWO we adopt a notional 6 m aperture, IWA $3\lambda/D$ and OWA $64\lambda/D$ at 0.76 μm, and continuum contrast at least 10^{-10}, following the LIFE X comparison (Carrión-González et al., 2023). This uses a gray-albedo approximation and reports geometry plus contrast access only, not spectral recovery.

4.4 A transmission-feature ceiling

For transiting planets we calculate an intentionally simplified high-mean-molecular-weight atmosphere with $\mu = 28$, $T = 255\overline{S}^{1/4}$ K and surface gravity from each mass–radius draw. A five-scale-height feature has amplitude

$$H = \frac{k_B T}{\mu m_u g}, \qquad \Delta_{\rm tr} \simeq \frac{2R_p(5H)}{R_*^2}. \tag{12}$$

This is a feature envelope, not a biosignature amplitude, and the chosen temperature is a scaling convention rather than a computed climate. A 4.3 μm bin of width 0.043 μm, collecting area 25.4 m^2, throughput 0.3 and observed K magnitude estimate photon noise, using a 666.7 Jy zero point at 2.159 μm with the local blackbody color correction (2MASS, 2008). Equal in- and out-of-transit photon budgets give the $\sqrt{2}$ noise factor. A circular duration with impact parameter 0.5 is used.

After 20 transits, we add residual floors of zero, 10 or 30 ppm in quadrature and compare feature attenuation factors 1 and 0.3. These floors and attenuation factors are scenarios, not universal properties of JWST. The output records $20P$ as an approximate cadence scale; twenty consecutive transit centers actually span $19P$, and visibility, observatory allocation and instrument scheduling add further constraints. Unknown transit masses completed with Eq. (8) can be especially misleading for volatile-rich planets, so these results are used to identify bottlenecks and reconnaissance targets, not to certify habitability.

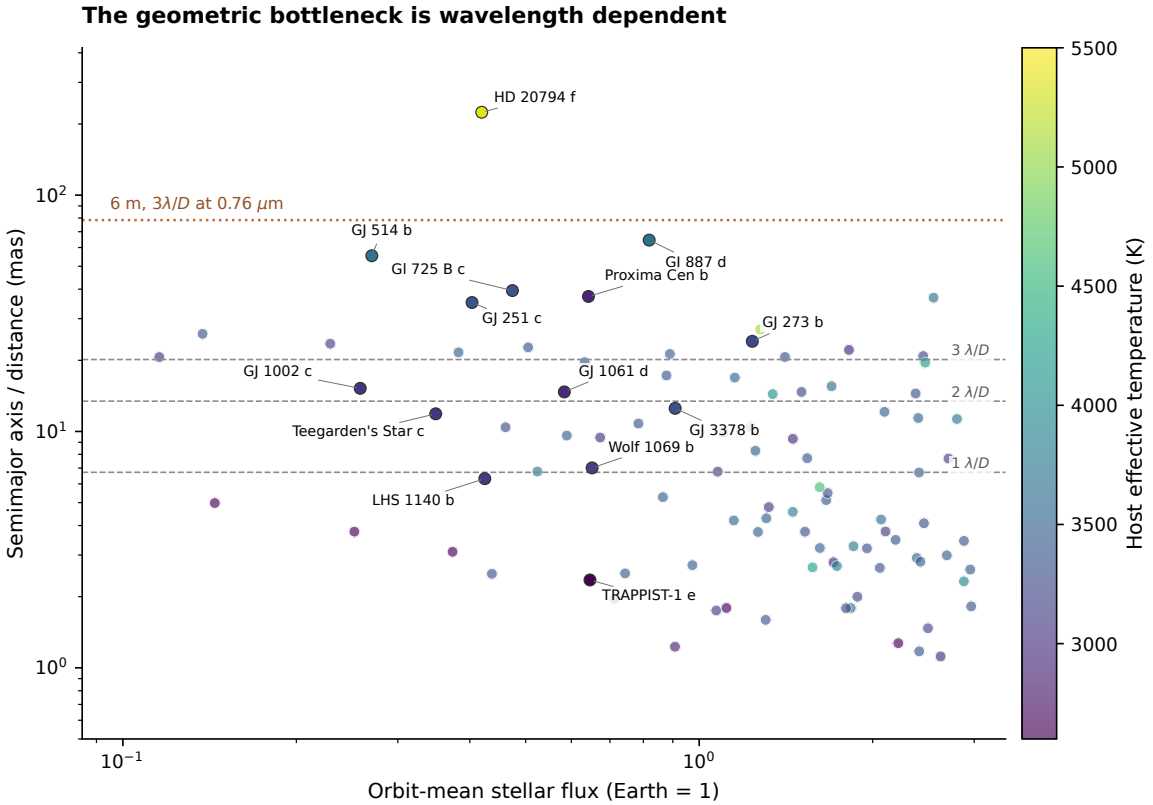


Figure 1: Audited planets within 80 pc in flux–angular-scale space. The ordinate is a/d; inclination, phase and eccentricity determine actual projected separation. Horizontal lines illustrate the wavelength-dependent angular bottleneck. Colors indicate stellar effective temperature, not habitability. HWO's line is a long-horizon geometry comparison.

5 Target results and their sensitivity

5.1 A nearby anchor and alternative habitat classes

Proxima Cen b is the strongest observational anchor in this scenario dictionary. It is first in 92.4% and in the first five in 99.2% of southern-telescope scenarios. The result expresses unusually favorable distance, angular scale and contrast combined with permissive compact-temperate states. It does not establish atmospheric survival around an active late-M star. Evolutionary models permit very different volatile and atmospheric outcomes depending on starting conditions (Ribas et al., 2016; Meadows et al., 2018).

GJ 887 d is the most consequential recently confirmed alternative in our accessible sample. At approximately 3.29 pc, its 0.212 AU orbit gives a nominal angular scale near 64.5 mas. Its minimum mass, about $6.1 M_{\oplus}$, weakens a narrowly terrestrial interpretation but allows a larger reflecting area in extended scenarios. The 3,688 K host also receives a different preference from late-M stars in the penalized cases. These facts make it useful as a test of a different habitat class, not a demonstrated rocky world. It is in the first five in 59.9% of the southern scenarios. Its discovery paper already identifies its temperate interest (Hartogh et al., 2026).

Gl 725 B c is the clearest example of an input-driven promotion that does not translate into a southern observing recommendation. The corrected 3.523 pc distance gives a 39.5 mas nominal separation instead of 20.3 mas. It ranks in the all-sky first five in 91.5% of our scenarios. However, its declination near

Table 3: Representative audited targets. Values are nominal catalog/override inputs; derived quantities are not new measurements.

Planet	d (pc)	$M/M_\oplus$	$R/R_\oplus$	$\overline{S}$	a/d (mas)	South
Proxima Cen b	1.30	1.05†	—	0.64	37.3	Yes
GJ 887 d	3.29	6.10†	—	0.82	64.5	Yes
Gl 725 B c	3.52	3.40†	—	0.47	39.5	No
GJ 273 b	3.79	2.89†	—	1.24	24.1	Yes
GJ 1061 d	3.67	1.67	—	0.58	14.7	Yes
GJ 251 c	5.58	3.88†	—	0.40	35.1	Yes
GJ 1002 c	4.85	1.36†	—	0.26	15.2	Yes
Teegarden's Star c	3.83	1.05†	—	0.35	11.9	Yes
Wolf 1061 c	4.31	3.41†	—	1.41	20.7	Yes
Ross 128 b	3.37	1.40†	—	1.51	14.7	Yes
GJ 3323 c	5.37	2.31†	—	0.23	23.5	Yes
TRAPPIST-1 e	12.43	0.69	0.92	0.65	2.4	Yes
TRAPPIST-1 f	12.43	1.04	1.04	0.37	3.1	Yes
LHS 1140 b	14.99	5.60	1.73	0.42	6.3	Yes
HD 20794 f	6.04	5.82†	—	0.42	224.1	Yes
GJ 514 b	7.62	5.20†	—	0.27	55.4	Yes

†: minimum mass $M \sin i$, not true mass. Dashes mark unmeasured quantities. South means culmination at least 30 deg at latitude −24.6 deg; it does not imply calendar or spectral feasibility. GJ 1061 d's true mass is conditional on the coplanar dynamical model. HWO comparison targets remain longer-horizon cases.

+59.6 deg gives a maximum altitude of only 5.8 deg at the adopted southern site. Its near-term roles are radio observations and supporting characterization; a northern future direct-imaging capability would need its own performance and schedule analysis. The discovery authors already recognize the planet's habitable-zone location (Ould-Elhkim et al., 2026).

GJ 1061 d is a complementary compact-world case, because the selected dynamical model constrains its true mass under coplanarity (Dreizler et al., 2025). Its nominal angular scale is only about 14.7 mas, close to the reference IWA, making performance assumptions decisive. GJ 251 c offers a larger angular scale but culminates at only about 32 deg at the southern site, just above our crude cutoff. A genuine observing proposal must treat that low elevation more carefully. GJ 1002 c and Teegarden's Star c provide cooler compact scenarios, yet small separations and lower reflected flux limit their scores. Good environmental arguments do not remove the geometric bottleneck.

5.2 A ranking is not a single durable ordering

Figure 2 makes several reversals visible. A permissive composition gate increases the role of GJ 887 d, while an empirical warm-edge interval restores GJ 273 b and Wolf 1061 c. A late-M penalty can change which of these physically different cases dominates. These reversals are informative: they identify the assumptions that a future measurement or a better population model would need to resolve.

The all-grid, equal-time minimax portfolio is GJ 887 d, Proxima Cen b, and Wolf 1061 c, with maximum fractional regret 46.0% relative to the scenario-specific three-target oracle. The solver reports a zero optimization gap over 35 candidates with nonzero utility. This mathematical result should be interpreted as an audit of the scenario set: the warm-edge member protects against empirical-zone scenarios, even though it is weak under the conservative interval. It is not a recommendation to allocate 300 actual telescope hours before validating atmosphere, albedo, contrast suppression and seasonal access. A conservative-zone-only calculation is also provided in the results package and discussed in Appendix A.

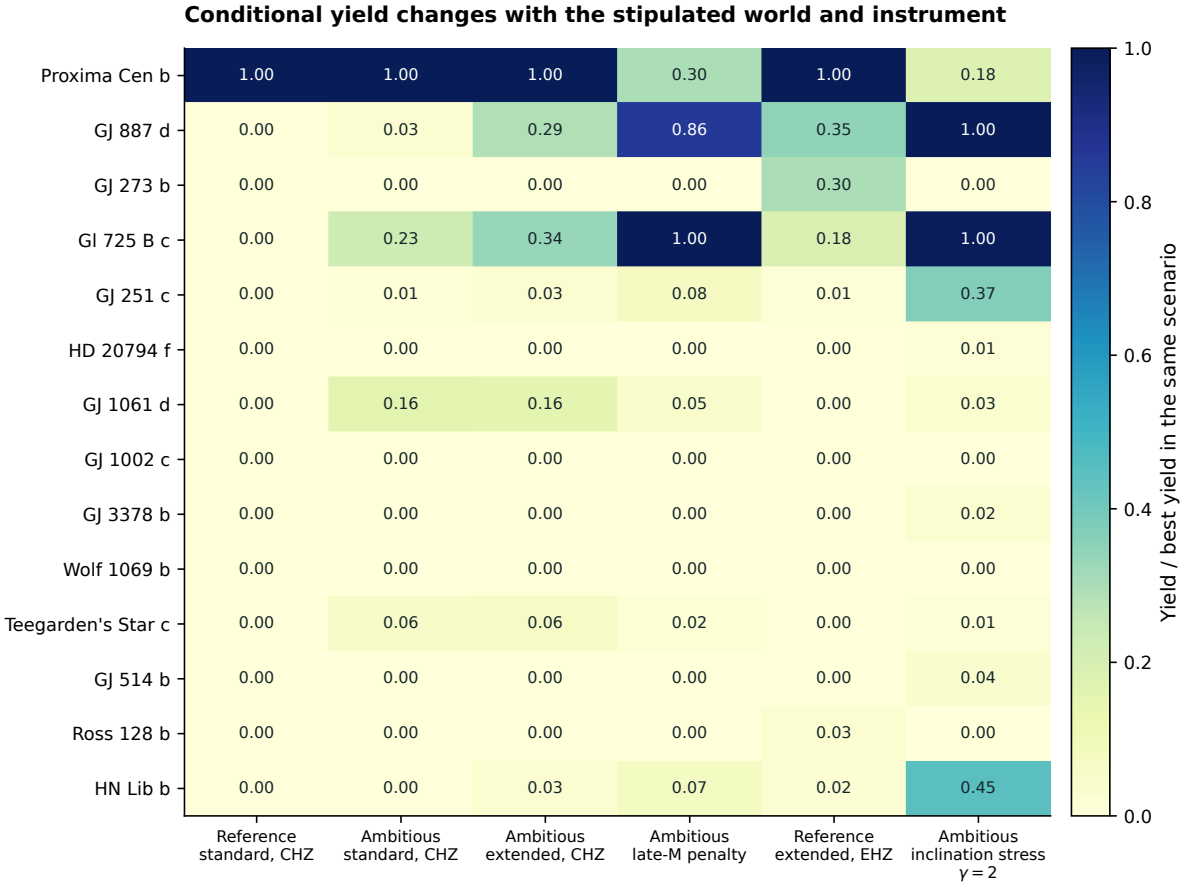


Figure 2: Selected all-sky scenarios, normalized separately to the best target in each column. CHZ and EHZ denote the conservative and empirical flux intervals. Cells compare the conditional weight in Eq. (2); neither the colors nor the ranks are probabilities of life. The all-sky view deliberately includes northern planets that fail the southern-site cut.

Table 4: Reference, standard-size, conservative-zone, orbit-mean comparison with compact radii and $\gamma = 1$. SE is numerical Monte Carlo error for the coupled expectation.

Planet	$\mathbb{E}[w]$	$\mathbb{E}[D]$	$\mathbb{E}[wD]$	$\mathbb{E}[w]\mathbb{E}[D]$	Ratio	$\mathrm{SE}(\mathcal{J})$
Proxima Cen b	0.978	0.428	0.408	0.419	1.0	2.65×10^{-3}
Gl 725 B c	0.702	0.0155	1.41×10^{-3}	0.0109	7.7	5.63×10^{-5}
GJ 887 d	0.0861	0.0413	1.83×10^{-4}	3.56×10^{-3}	19.4	2.00×10^{-5}
GJ 1061 d	1	2.69×10^{-5}	2.69×10^{-5}	2.69×10^{-5}	1.0	6.29×10^{-7}
GJ 251 c	0.581	1.67×10^{-3}	4.51×10^{-6}	9.68×10^{-4}	214.7	1.35×10^{-7}
GJ 1002 c	0.924	2.54×10^{-4}	1.21×10^{-6}	2.34×10^{-4}	193.3	3.18×10^{-8}

All-sky values are shown, so Gl 725 B c is retained even though the southern-site score is zero. Very large ratios at tiny absolute scores are diagnostics, not strong discovery prospects.

5.3 Coupling errors are large for some attractive targets

The baseline values in Table 4 illustrate why marginal habitability and observability cannot simply be multiplied. For GJ 887 d the product approximation is 19.4 times the coupled score, and for Gl 725 B c it is 7.7 times the coupled score. The low-inclination, large-radius states that help marginal detectability are often the same states that fail a terrestrial mass preference. Proxima's correction is much smaller.

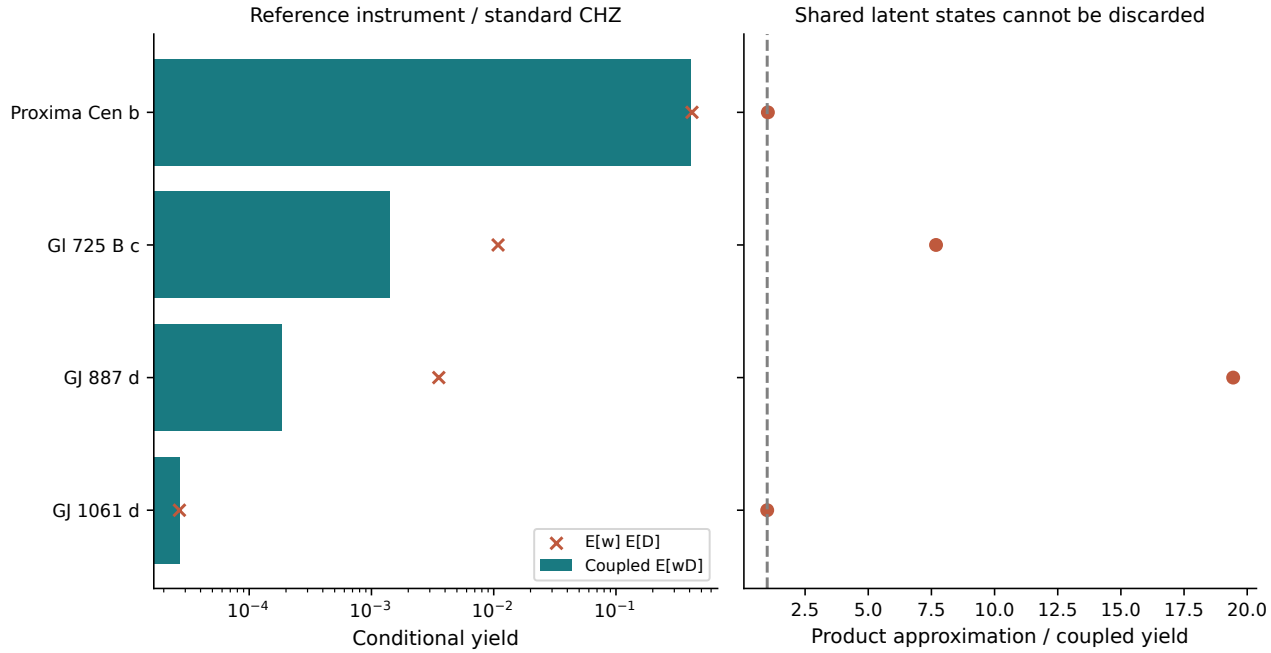


Figure 3: The product of marginal averages versus the coupled expectation in the reference instrument, standard-size, conservative-zone, orbit-mean scenario. All-sky values are shown. The right panel is a ratio diagnostic; confidence intervals for the paired covariance and the joint estimator are supplied in the numerical tables. Ratios involving very small denominators require particular care.

Table 5: Effect of the documented catalog-parameter policy in the baseline scenario. Both runs use 8,192 draws, 128 phases and the same base seed.

Planet	Raw $\mathbb{E}[w]$	Audited $\mathbb{E}[w]$	Raw $\mathcal{J}$	Audited $\mathcal{J}$
GJ 273 b	0.26	0	6.02×10^{-5}	0
Gl 725 B c	0.702	0.702	1.91×10^{-5}	1.41×10^{-3}
Wolf 1061 c	0.231	0	1.34×10^{-3}	0
GJ 3323 c	0.277	0.345	5.67×10^{-7}	6.56×10^{-7}

Zero denotes no eligible recovery in this finite simulation under this gate; it is not a zero probability of life. This joint audit changes every documented parameter for each host, so it is not a distance-only experiment. Source policies are retained in the supplement.

The sign and size of the effect are target- and scenario-dependent; covariance is not universally negative.

Neither the product approximation nor the corrected score is a life probability. Correcting this statistical error improves internal consistency, but cannot repair a wrong climate gate, a wrong radius completion, or an unrealistic instrument floor. Paired Monte Carlo uncertainties are reported for the difference using physical-state draws as the independent sampling units.

5.4 The stellar-data ablation

The raw-versus-audited experiment in Table 5 changes only the documented parameter policy under otherwise matching simulation choices. It separates two mechanisms that a naive uncertainty ranking can conflate. A corrected distance can improve angular access. A more precise luminosity can *remove* apparently favorable eligibility that arose from an unrealistically broad or superseded distribution. The Wolf 1061 and GJ 273 examples show why a large uncertainty tail is not positive evidence for a habitable

Table 6: Illustrative high-μ, five-scale-height envelopes with a 10 ppm residual floor and no feature attenuation. The noise is per effective 4.3 μm bin.

Planet	Feature (ppm)	Noise, 20 transits (ppm)	Duration (h)	$20P$ (yr)
TRAPPIST-1 e	72.5	34.1	0.76	0.33
TRAPPIST-1 f	61.5	32.0	0.88	0.50
TRAPPIST-1 g	55.3	30.7	0.97	0.68
LHS 1140 b	16.3	15.6	1.74	1.35
TOI-700 d	5.3	13.4	2.88	2.05
TOI-700 e	6.8	13.6	2.63	1.52

No atmospheric presence or biosignature is assumed to have been established. $20P$ is an approximate cadence scale; visibility and allocated telescope time are additional constraints. The spectrum-injection experiment uses a separate achieved-noise grid.

state. Coherent measurements can reduce a score while improving the quality of the decision.

5.5 Transmission reconnaissance and the longer-horizon comparison

The transmission ceiling supports TRAPPIST-1 e/f and LHS 1140 b as atmospheric reconnaissance cases, while showing why a feature-scale calculation cannot establish that a biological attribution is feasible. Table 6 reports selected envelopes and the modeled 20-transit noise. The assumed scale height, clouds, mass completion and stellar residuals all matter. TOI-700 d/e are useful environmental comparisons, but their larger host and small atmospheric depth impose a strong signal penalty in this calculation.

Current observations justify caution in a specific, evidential way. Four JWST PRISM visits of TRAPPIST-1 e have variable stellar contamination and permit both bare-rock and secondary-atmosphere interpretations (Espinoza et al., 2025; Glidden et al., 2025). Five NIRISS visits of f contain flares; the reported absence of detectable unocculted-heterogeneity contamination in those visits does not guarantee universally clean spectra, and high-mean-molecular-weight atmospheric limits depend on reduction choices (Lim et al., 2026). Differential observations in a shared system can therefore be valuable for learning about stellar nuisance and atmosphere retention even before life attribution becomes possible.

LHS 1140 b has measured bulk properties compatible with several interior and envelope interpretations, with tentative atmospheric inferences rather than a confirmed nitrogen atmosphere (Cadieux et al., 2024a,b). A 2026 ground-based helium report (Cherubim et al., 2026) is challenged by analyses of four noncontemporaneous JWST visits (Bennett et al., 2026; Radica, 2026). Those reanalyses share the same visits and must not be counted as independent observing campaigns. Helium nondetection is not a demonstration that no atmosphere exists. This makes the target useful for resolving an atmospheric question, not evidence that life is present.

For the 6 m, 0.76 μm geometry comparison, HD 20794 f—the 647.6-day planet called d in Nari et al. (2025)—has modeled access 55.1%, far above GJ 514 b at 8.9%. Its nearby G-star host and larger angular separation merit longer-horizon characterization. However, minimum mass near $5.8M_{\oplus}$, unknown radius and eccentricity near 0.45 leave composition and climate unresolved. An orbit-mean habitable-zone position is not proof of a stable surface ocean; the residence criterion and a dedicated thermal-inertia model ask different questions. This case is not promoted as a near-future HWO observation.

6 Simulated inhabited atmospheres versus observed spectra

The catalog surrogate assigns a feature to each eligible world, but a living planet need not produce that feature. To test this failure directly we use published, physically calculated transmission spectra

rather than an invented oxygen/methane cartoon. The experiment reuses a finite set from Eager-Nash et al. (2024a), archived by the authors at Zenodo (Eager-Nash et al., 2024b). It is a new observation-and-classification experiment on existing atmospheric calculations; we do not claim to have run a new photochemical or general circulation model.

6.1 Physical templates and observation model

The archive contains ten PSG transmission spectra: four abiotic atmospheres, four ecosystem-1 inhabited models at hydrogen outgassing rates of 0.1, 1, 10 and 100 Tmol yr^{-1}, and two ecosystem-2 variants at 0.1 and 1 Tmol yr^{-1} that include oxygen-consuming metabolisms. The code labels the latter `oxic`, but this does not mean a modern oxygen-rich atmosphere. The source calculations use a prescribed common early-Earth climate profile, a one-dimensional atmosphere and no cloud/haze treatment, with historical model parameters for TRAPPIST-1 e. They do not establish which of those states the actual planet occupies.

The headerless deposited total atmospheric transit contributions are fractional depths; multiplying by 10^6 gives ppm, despite a saved `GENERATOR-RADUNITS=ppm` configuration flag. The approximately 0.004922 solid-disk depth is not in that column and is removed as a continuum nuisance in the experiment. The units were checked against the source's Figure 17 plus the configured solid-disk baseline, rather than trusting the inconsistent units flag. We integrate the already $R = 200$ spectra into 184 logarithmic bins between 0.8 and 5 μm, corresponding to $R \simeq 100.4$, with a flat within-bin response.

For wavelength bins k, l the declared covariance is

$$K_{kl} = \sigma_w^2 \delta_{kl} + \sigma_c^2 \exp\left[-\frac{|\ln(\lambda_k/\lambda_l)|}{0.05}\right]. \tag{13}$$

We test $\sigma_w = 2, 5, 10, 20, 30, 50$ ppm and $\sigma_c = 0, 5$ ppm. These are *achieved measurement scenarios*, not a claim that a given number of JWST visits reaches those precisions across the band. A free constant and a slope in log wavelength are projected out before whitening. Known covariance and an exactly specified finite template family make this an optimistic discrimination experiment.

Four spotted-star abiotic variants use an unocculted 3% blackbody spot fraction at 2,300 K against a 2,566 K photosphere. Airless continua with and without the same spots are also included. This is a continuum stress test; stellar molecular lines, flares and evolving heterogeneity require richer models. The admitted null therefore has ten members: four abiotic atmospheres, four spotted variants, and two airless cases.

6.2 Finite-family likelihood and calibration

For a measured spectrum y, the test statistic is

$$T(y) = \log\left[\frac{1}{6}\sum_{b\in\mathcal{B}} p(y \mid b)\right] - \log\left[\frac{1}{10}\sum_{a\in\mathcal{A}} p(y \mid a)\right]. \tag{14}$$

Here $p(y \mid \text{template})$ is the density of the continuum-projected spectrum: for an orthonormal projection Q removing constant and slope, the mean is $Q^\mathsf{T}\mu$ and covariance is $Q^\mathsf{T}KQ$. Equal weights are computational choices within each model family, not frequencies of biological ecosystems. The threshold is the largest empirical 99th percentile obtained from 20,000 independent calibration draws for each admitted null. A separate 20,000 draws per injected state estimate recovery and false-positive rates. Wilson intervals quantify test-sample binomial uncertainty, conditional on the fitted threshold; they omit calibration and model uncertainty. The nominal 1% target is not a finite-sample guarantee.

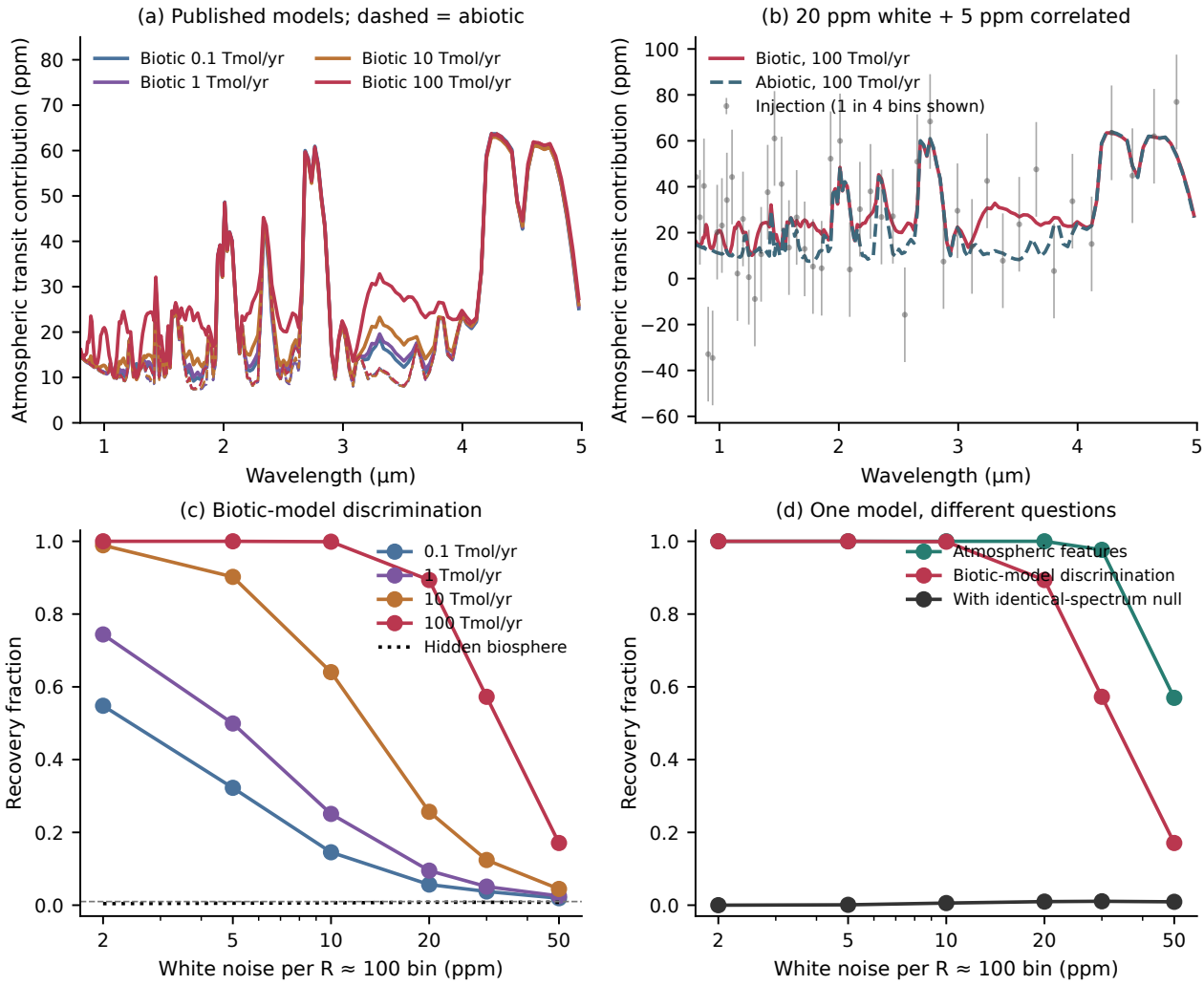


Figure 4: Published atmospheric models under a new declared observation experiment. (a) Six inhabited and four abiotic source spectra are used; the panel shows the four ecosystem-1 pairs. (b) One noisy realization; only every fourth bin is plotted. (c) Biotic-family recovery at 5 ppm correlated noise, with a hidden-life counterfactual. (d) Atmospheric-feature recovery, discrimination within the admitted finite families, and discrimination when an identical-spectrum null is admitted. The precision axis is not an exposure-time forecast.

We also test an atmospheric-feature question against the two airless alternatives. This is separate from biotic-family discrimination. To expose nonidentifiability we add two counterfactual diagnostics: an inhabited but spectrally hidden state with an abiotic template, and an abiotic-labeled state with exactly the spectrum of the strongest inhabited case. The latter is a mathematical adversary, not a newly discovered abiotic production mechanism. If life and no-life hypotheses give identical distributions, every decision rule satisfies TPR = FPR; increasing collecting area cannot remove that equality.

6.3 A living atmosphere can be easy to see and hard to identify

At $\sigma_w = 20$ ppm and $\sigma_c = 5$ ppm, all six inhabited atmospheres show atmospheric-feature recovery in all 20,000 held-out trials. Yet the corresponding biotic-family recoveries range from 2.99% to 89.355% (Table 7). The result demonstrates a difference between questions, not an observed atmospheric detection on TRAPPIST-1 e.

Table 7: Independent held-out recovery at 20 ppm white plus 5 ppm correlated noise. Intervals are 95% Wilson intervals for 20,000 trials and condition on the calibrated threshold and model family. Ecosystem 2 includes oxygen-consuming metabolisms.

H_2 outgassing	Inhabited state	Recovery (%)	95% interval (%)
0.1 Tmol yr^{-1}	Ecosystem 1	5.665	5.35–5.99
0.1 Tmol yr^{-1}	Ecosystem 2	2.990	2.76–3.24
1 Tmol yr^{-1}	Ecosystem 1	9.495	9.10–9.91
1 Tmol yr^{-1}	Ecosystem 2	6.135	5.81–6.48
10 Tmol yr^{-1}	Ecosystem 1	25.660	25.06–26.27
100 Tmol yr^{-1}	Ecosystem 1	89.355	88.92–89.77
Hidden-life counterfactual (abiotic-equivalent spectrum)		0.810	0.69–0.94
Strongest state with identical-spectrum null admitted		0.980	0.85–1.13

The strongest source state is recovered in 89.355% of trials against the admitted ten-model null, falling to 0.98% after adding the identical-spectrum alternative. Its atmospheric features remain visible. The source authors' own twenty-transit NIRSpec/PRISM PandExo experiment did not robustly recover the modeled methane features (Eager-Nash et al., 2024a); our uniform achieved-noise experiment is not a revision of that instrument forecast. The hidden-life state gives only 0.81% apparent biotic recovery. Such a state is not physically predicted by this counterfactual construction, but spectrally cryptic biospheres are independently motivated by Earth-history modeling (Reinhard et al., 2017). A biological population model that assigns every inhabited world a modern-Earth-like observable atmosphere would omit an important class of false negatives.

Even 2 ppm white noise with the assumed 5 ppm correlated component recovers the weak ecosystem-2, 0.1 Tmol yr^{-1} state in only 11.795% of trials. More precise white-noise measurements do not necessarily solve a correlated-shape or model-degeneracy problem. Conversely, the strong source state is nearly always recovered at sufficiently low noise within this finite null. Both outcomes matter; the experiment is not designed to conclude that all atmospheric inference is impossible.

6.4 A missing null can dominate an apparently decisive result

An initial null set containing only the eight atmospheric abiotic/spot models gives a 99.53% biotic-call rate for a flat spectrum and 97.495% for a spotted airless continuum at 20+5 ppm. Adding the two missing airless alternatives reduces those rates to 0.035% and 0.030%, while strongest-state biotic recovery remains 89.355%. The apparent certainty came from comparing several wrong atmospheric explanations: the biotic family could be relatively closer to a flat spectrum even though the planet had no atmosphere in the injection. This failure is retained as a named diagnostic rather than suppressed after correcting the main analysis.

The largest false-positive rate among the ten admitted nulls is 1.125%. Other omissions are not uniformly innocuous, but are much smaller than the airless failure. Omitting the highest-outgassing abiotic family gives 1.285% at 10+5 ppm and 1.195% at 20+5 ppm. Test-only Wilson intervals for those cases exclude 1%, but calibration uncertainty and multiple comparisons are not included. The appropriate inference is modest miscalibration in those tested omissions, not proof that the untested abiotic space is safe. Likelihood ratios should be accompanied by checks of absolute model adequacy and an explicit accounting of missing hypotheses.

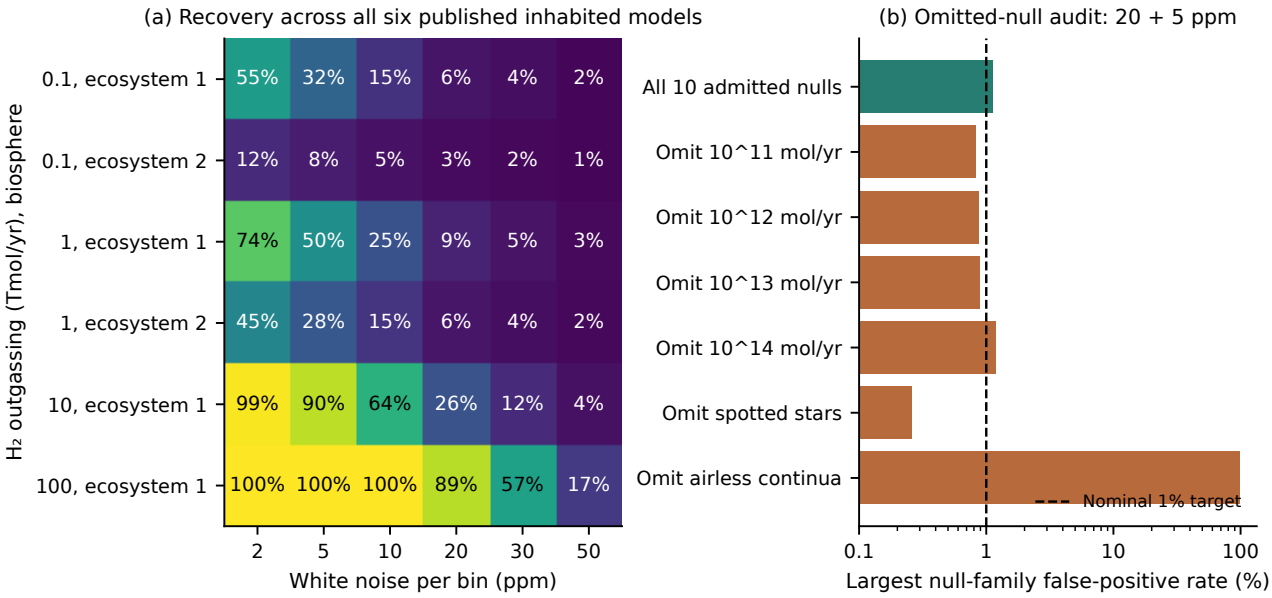


Figure 5: Recovery and model-misspecification audit. Left: all six inhabited source states at 5 ppm correlated noise. Right: the largest false-positive rate for each held-out null family at 20+5 ppm. Removing airless continua creates a catastrophic failure; removing the tested geological or spotted families gives much smaller excesses. The main analysis includes both airless alternatives. The finite admitted null is still not exhaustive.

7 Technosignatures: targets, power assumptions, and null information

Technology does not need to reside on the catalog planet that supplied a system's discovery label. A civilization could use another planet, an orbital platform, or an artifact; the probability of such behavior is unknown. We therefore run the radio screen over all positive-distance, noncontroversial confirmed-planet hosts, not just the atmospheric sample. This yields 4,792 unique host names. It is a controlled targeting universe, not a claim that confirmed-planet hosts should be favored over equally nearby stars lacking known planets.

We restrict numerical technosignature simulations to unresolved narrowband radio emission. Atmospheric pollutants, optical pulses, artificial illumination and waste heat have different signal and background models; an Earth-technology comparison demonstrates how widely their detectable distances differ (Sheikh et al., 2025). Our outputs are consequently radio-family priorities, not an all-technologies civilization ranking.

7.1 A transparent radio sensitivity comparison

For an unresolved line, the minimum equivalent isotropic radiated power is

$$P_{\min} = 4\pi d^2 \, q_r \, \mathrm{SEFD} \, 10^{-26} \sqrt{\frac{\Delta\nu}{n_{\mathrm{pol}} t}}, \tag{15}$$

with SEFD in Jy, distance in meters, channel width $\Delta\nu$ in Hz and dwell time t in seconds. The intrinsic linewidth cancels when it is narrower than the matched channel; a broadband leakage signal cannot use this formula unchanged (Price et al., 2020; Tusay et al., 2024). For SEFD=10 Jy, $q_r = 10$, $\Delta\nu = 1$ Hz, two polarizations and $t = 300$ s,

$$P_{\min} = 4.88467 \times 10^8 (d/\mathrm{pc})^2 \ \mathrm{W}. \tag{16}$$

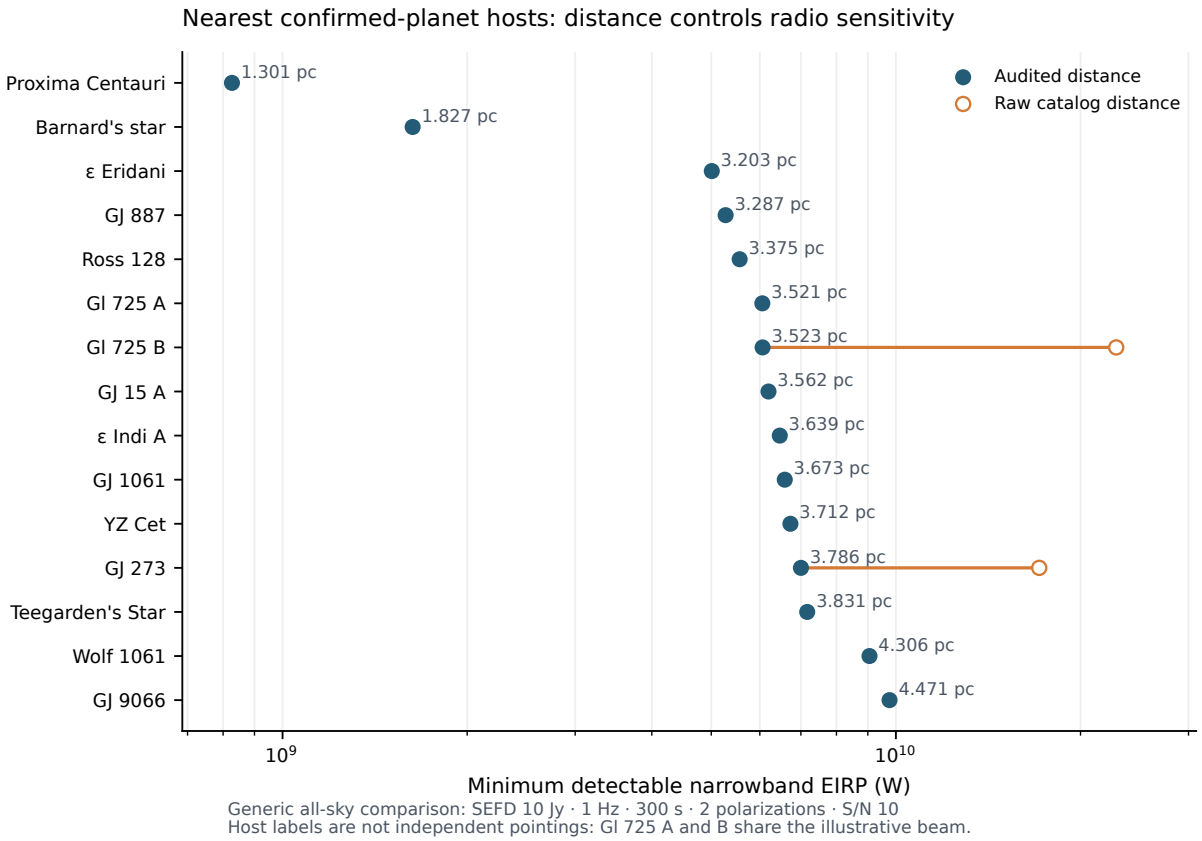


Figure 6: Illustrative narrowband source-power thresholds for nearby confirmed-planet hosts, with distance-audit changes retained. Apparent radio attractiveness is conditional on the searched signal being emitted toward Earth at an observed frequency and time. The model is not an actual all-sky instrument or a new SETI observation.

This generic all-sky comparison is grounded in a published receiver benchmark. It is not a claim that one telescope sees every host with identical sensitivity. A 34 Jy column illustrates linear SEFD scaling. EIRP is beam-equivalent power, not total civilization energy consumption.

Proxima's threshold is 8.27×10^8 W, Barnard's star's is 1.63×10^9 W, and those of ϵ Eridani and GJ 887 are about 5.0–5.3×10^9 W. These last systems illustrate why a system radio target need not have the highest score for contemporary terrestrial surface habitability. The corrected distances reduce Gl 725 B's threshold by a factor 3.77 and GJ 273's by 2.45. No probability of local life follows from these received-flux comparisons.

7.2 Mock inhabited systems with explicit emitters

Conditional on a system producing a transmitter of the modeled family, let

$$x = \log_{10}(P/\mathrm{W}), \quad 7 \le x \le 15, \qquad p(x \mid \beta) \propto 10^{-\beta(x-7)}, \quad \beta = 0, 1, 2. \tag{17}$$

Thus $\beta = 0$ is log-flat and $\mathrm{d}N/\mathrm{d}P \propto P^{-(\beta+1)}$. These are illustrative luminosity distributions, not measured extraterrestrial priors. A Gaussian matched-filter response gives $p_{\mathrm{pass}}(P) = \Phi_N[q_r(P/P_{\mathrm{min}} - 1)]$. We apply a pipeline-recovery factor $r = 0.94$, motivated by a published injection experiment but not assumed universal across facilities (Margot et al., 2023).

If the transmitter is on with duty fraction u, visits sample independent temporal states, and a

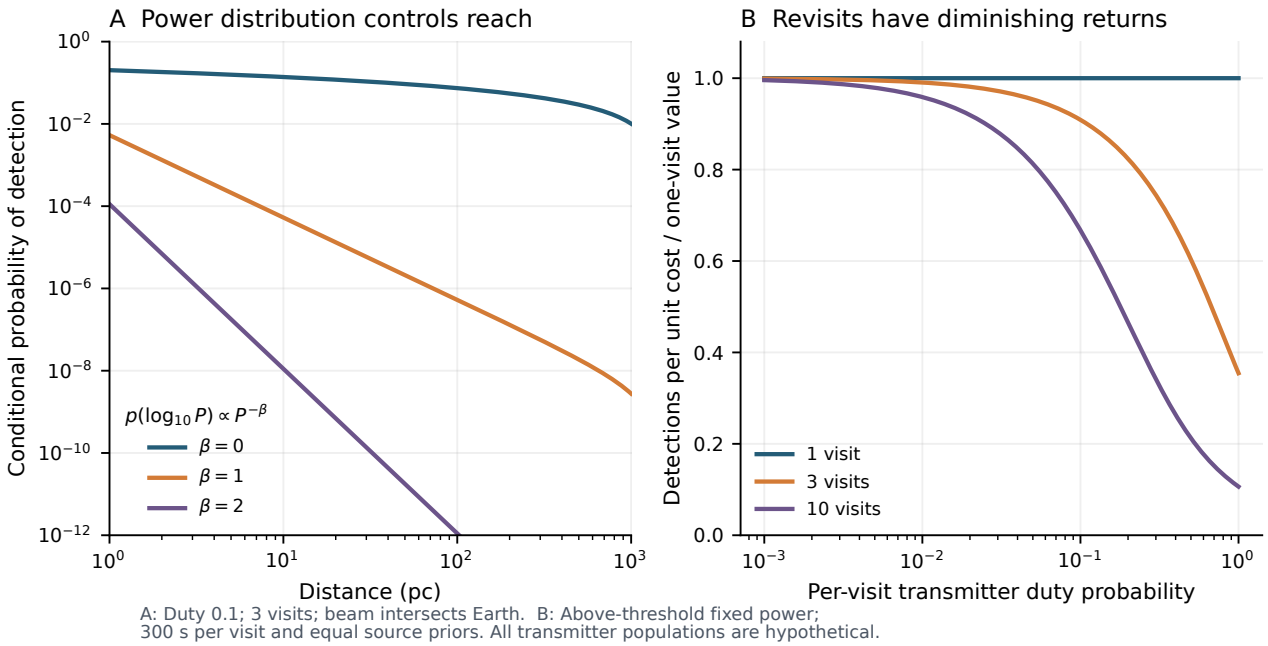


Figure 7: Dependence of conditional radio recovery on the assumed power population and temporal sampling. The revisiting comparison assumes independent duty states, fixed power and beam geometry, equivalent target priors, and no overheads. Real confirmation requirements and cadence correlations can alter the allocation tradeoff.

persistent beam intersects Earth with probability b, completeness after n visits is

$$c_s = b \int p(x \mid \beta) \left[1 - (1 - ur\, p_{\text{pass}}(10^x))^n\right] \mathrm{d}x. \tag{18}$$

Power and beam direction remain fixed across revisits. Putting b inside the exponent would incorrectly resample the beam direction at each epoch. We test duty fractions 0.01, 0.1 and 1; one, three or ten visits; and beam fractions 1, 10^{-2} and 10^{-4}. Frequency coverage, Doppler-search bounds, residual interference, signal lifetime and the retarded emission epoch remain additional conditioning variables. We do not infer that a civilization exists "now" from a signal emitted a light-travel time d/c earlier.

For three 300-second visits at duty 0.1, conditional on the beam crossing Earth, Proxima's completeness is 0.195 for the log-flat power distribution, 0.00315 for $\beta = 1$, and 3.91×10^{-5} for $\beta = 2$. The approximately 5,000-fold range comes from the stipulated emitter population, without changing the telescope or host. There is no evidential basis for presenting the first value alone as the detection chance for an actual civilization.

Numerical integration is checked against one million importance-sampling draws over 540 nearby-host scenarios: twenty hosts, three power slopes, three duties and three visit counts. The largest quadrature–Monte Carlo discrepancy is 3.40 Monte Carlo standard errors. The full host table retains thresholds; the expensive population simulation is restricted to the nearest twenty labels. Other checks reproduce radiometer scaling, normalized power densities, Gaussian amplitude recovery and the fixed-beam revisit formula.

7.3 The same null observations can be informative or nearly empty

The nearest twenty host names occupy nineteen illustrative 9-arcmin-FWHM pointing groups because Gl 725 A and B share a beam. Sharing a beam saves observing time; it does not prove that host occurrence states are dependent. We calculate both twenty independent-host occurrence trials and a nineteen-group

Table 8: Targets promoted by a stated scientific role. These are research priorities, not approved observing proposals or inferred inhabited-world probabilities.

Role	Targets	Measurement or assumption that controls the next decision
Near-future reflected-light anchor	Proxima Cen b	Validate achievable contrast and molecular-template recovery; constrain atmospheric and stellar alternatives.
Alternative temperate habitat	GJ 887 d	Radius/composition, inclination, atmospheric survival and realistic phase-dependent spectrum.
Compact-world comparison	GJ 1061 d; conditional GJ 1002 c and Teegarden c	Small-IWA performance, ephemeris knowledge, radius completion and cool-climate atmospheric requirements.
Northern opportunity	Gl 725 B c; GJ 251 c	Correct source parameters, northern facility performance and actual elevation/scheduling; neither is an automatic southern recommendation.
Warm-edge controls	GJ 273 b; Wolf 1061 c; Ross 128 b	Climate and albedo can change eligibility and recovery together; useful tests of hot-edge atmosphere models.
Transmission reconnaissance	TRAPPIST-1 e/f; LHS 1140 b	First resolve atmosphere and stellar-contamination alternatives; evaluate discriminatory spectral information rather than a single line.
Longer-horizon direct imaging	HD 20794 f (647.6-day planet); secondary GJ 514 b	HWO architecture plus true mass, radius, eccentric climate and wavelength-dependent contrast.
Near-term radio coverage	Nearest confirmed-planet systems, including Proxima, Barnard's star, ϵ Eridani, GJ 887 and Gl 725 A/B	Search-frequency coverage, power distribution, beaming, duty correlation, recovery calibration and shared pointing cost.

alternative with one latent occurrence per group. Both use nineteen pointings and three visits, totaling 4.75 hours on source before overheads.

For a specified transmitter-family occurrence fraction η, hypothetical all-null data have likelihood

$$\mathcal{L}(\eta \mid 0) = \prod_s (1 - \eta c_s). \tag{19}$$

The product is conditional on independence of the modeled occurrence units and recovery events. A uniform prior on $\eta \in [0, 1]$ is used solely for the illustration. This treatment is related to existing Bayesian SETI inference, not a new general theory (Grimaldi & Marcy, 2018; Margot et al., 2023).

With beam fraction and duty both one, the hypothetical log-flat population yields a 95% upper bound of 0.203 under independent hosts, or 0.212 under group occurrence. With $\beta = 1$, the bounds are about 0.949 in either model, almost the prior's own 0.95 quantile. At duty 0.1 even the log-flat independent-host bound weakens to 0.711. These are *synthetic* nondetections. They demonstrate that a limit on a specified emitter family can range from informative to nearly uninformative without changing the observed null outcome. They are not limits on civilization prevalence.

8 A practical, differentiated target program

The strongest recommendation is a program with different scientific jobs, rather than one universal descending list. Table 8 summarizes the decisions supported by this analysis. None requires pretending that a life prior has been measured.

The unusual cases here are not presented as discoveries of previously unknown habitable planets. GJ 887 d is newly relevant to a confirmed-only comparison and changes the composition tradeoff. Gl 725 B c is unusually sensitive to a catalog distance error and to hemisphere. GJ 273 b becomes easier to

resolve while losing conservative-zone support under better stellar data. Those are reproducible changes in inference, which are more informative than an unsupported declaration that a planet was missed by the field.

Atmospheric follow-up should first reduce uncertainties that can reverse a decision. For example, a reliable radius or inclination for GJ 887 d addresses a different risk from another small improvement in its orbital period. Better stellar and atmospheric models for TRAPPIST-1 can improve interpretation across several planets, but shared nuisance does not make their atmospheric outcomes identical. A quiet present-day optical light curve is not a measurement of integrated XUV exposure or initial volatile inventory; neither a late-M veto nor an age-based civilization bonus is justified as a universal empirical rule.

Cross-channel observing can be economical without conflating evidence. Radio observations of a planetary system can continue while its atmosphere remains uncertain, and an atmospheric null does not exclude an orbital technological source. Conversely, a nearby giant-planet system can be a good radio pointing without becoming a terrestrial biosignature candidate. The common target name is a coordination opportunity, not a common probability model.

9 Verification, limitations, and reproducibility

9.1 Checks that were actually performed

The orbital engine reproduces the exact time-averaged eccentric flux in Eq. (5) and the isotropic circular-orbit access probability $\sqrt{1-(\mathrm{IWA}/(a/d))^2}$ when the IWA is smaller than a/d. The covariance identity holds numerically, and shuffling shared states removes the constructed covariance. The inclination proposal reproduces known normalization and mean-cosine moments. A higher-resolution independent-seed run uses 16,384 physical draws and 256 phases for eighteen selected targets. Appendix A reports the resulting changes, alongside the eccentricity and raw-catalog ablations.

The spectral engine preserves constant and linear functions under rebinning, reconstructs the sufficient likelihood subspace to numerical precision, and agrees with an independent two-simple-Gaussian-hypothesis result. For the selected physical template pair at 20+5 ppm, analytic power is 91.937% and simulated power is 92.074%; simulated false-positive rate is 1.023% against 1%. This validates the likelihood implementation, not the truth of the atmospheric models. The strongest practical review correction was inclusion of the previously missing airless alternatives, whose failure is explicitly retained in Figure 5.

9.2 What would be needed for a stronger claim

There are five principal limitations. First, the environmental preferences are not biological priors. Composition, surface water, atmospheric retention and abiogenesis are incompletely measured, and the scenario set is not exhaustive. Subsurface biospheres, exomoons, free-floating planets, non-water solvents, deliberately engineered habitats and many volatile-rich climates are outside the atmospheric screen. A low score is not evidence of sterility.

Second, the physical uncertainty distributions combine published marginals and explicit defaults. They do not include all stellar–planetary covariances, all original radial-velocity activity models, or a planet-existence posterior. Source corrections are reproducible but not a replacement for re-fitting raw data. Nominal screening cuts can exclude plausible tails, and the radius completions do not establish composition.

Third, the ground-based and transit calculations are surrogates. Correlated albedo, clouds, surface temperature and feature depths are not solved together; a higher reflected continuum can coexist

with weaker molecular features. The raw-leakage extrapolation, residual floors, constant equivalent feature and idealized phase coaddition need facility-specific validation. The current calculations do not include telluric line confusion, local sky background, exozodiacal light, detector systematics, stellar line variability, calendar access or competition for time. Some omitted noise sources are decisive precisely where the toy model is most optimistic.

Fourth, the atmospheric injection test spans a finite source grid with known noise covariance. It measures discrimination among admitted models, not posterior life probability. Its airless failure demonstrates why a large relative likelihood does not establish model adequacy. The exact-spectrum adversary is a theorem illustrated numerically, not a physically established abiotic explanation for the source's strongest biosphere. Full climate/photochemistry retrievals, broader abiotic models, stellar spectral libraries and independent validation observations would be needed for a claim of biological attribution.

Fifth, the radio completeness is conditional on a signal class, power distribution, duty model, persistent beam intersection and effective searched frequency. These are not inferred from human technological history as though humanity were a random sample of civilizations. Shared pointing cost is separated from host occurrence independence, but a real survey also needs coupled interference and recovery models. The finite list of confirmed systems is not necessarily the optimal all-star radio survey universe.

9.3 Data and code availability

The accompanying reproducibility archive contains the unmodified dated catalog; the query and checksum manifest; all fifteen parameter overrides and their source table; the public spectral archive and its verified hashes; executable simulation and table-generation code; numerical results, diagnostics and vector figures; and the LaTeX source. The principal programs are `target_audit.py`, `spectral_injections.py`, and `radio_audit.py`. Seeds, sample counts, software versions and exact run commands are documented in `README.txt`. Intermediate copies of literature PDFs are not required and are not redistributed. All observing results in this paper are simulated; no new telescope data were collected.

10 Conclusions

The strongest defensible result is a conditional, auditable observing program. Proxima Cen b remains a robust nearby reflected-light anchor within the tested assumptions. GJ 887 d is a useful alternative when larger temperate worlds are admitted. Gl 725 B c is promoted by a corrected distance but excluded from the southern telescope by sky position. GJ 1061 d tests a compact-world case close to the angular limit; warm-edge targets remain useful controls rather than secure habitat winners. TRAPPIST-1 e/f and LHS 1140 b are priorities for resolving atmospheric and stellar alternatives, while HD 20794's 647.6-day planet becomes compelling in a longer-horizon geometry comparison.

Three methodological findings are more durable than the precise ordering. Shared physical states can make a product of separately averaged eligibility and observability seriously overoptimistic. Detecting an inhabited atmosphere's features can be much easier than attributing them to life, and a missing airless null can create an almost universal false positive. Radio nondetections constrain a specified emitter population only when the survey is sensitive to that population. These findings identify concrete work that can improve the next target decision without manufacturing confidence about where life or civilization actually exists.

Preparation and attribution

This research manuscript and its original computational experiments were prepared with AI assistance in response to a research brief. Published atmospheric models and public catalog data remain attributed to their original authors and providers. The work has not been submitted to arXiv or peer reviewed. No human authorship, institutional affiliation or observational campaign is asserted.

A Further robustness results

A.1 Numerical convergence

The independent run uses seed 20261117, 16,384 physical draws and 256 phase nodes for eighteen selected targets, compared with seed 20261017, 8,192 draws and 128 nodes in the main run. Among 11,664 matched scenario rows, the largest absolute change in $\mathcal{J}$ is 0.0134. The largest difference divided by the quadrature sum of numerical Monte Carlo errors is 3.07; no matched row exceeds five such errors. These comparisons are correlated across scenarios and are not 11,664 independent statistical tests. They check physical-state Monte Carlo error and a simultaneous increase in phase-grid resolution; they do not validate physical assumptions.

Replacing the eighteen main-run rows by their higher-resolution counterparts changes the largest top-five scenario frequency by 1.85 percentage points. Proxima's first-place frequency changes from 92.44% to 92.13%. The minimax portfolio becomes GJ 887 d, Proxima Cen b, Wolf 1061 c with maximum regret 45.17%. Candidates not in the validation subset retain main-run draws in this comparison. Low absolute-score ordering remains more fragile than the leading target conclusion.

A.2 Dependence on the environmental decision set

Restricting the optimization to the 324 conservative-zone scenarios gives GJ 251 c, GJ 887 d, Proxima Cen b with maximum regret 23.72% and mean regret 3.91%. The all-grid optimum instead contains a warm-edge target because empirical-zone scenarios remain in its adversarial set. Thus portfolio membership can express a decision maker's environmental scope rather than numerical evidence for life. Both optimizations have zero reported mixed-integer optimality gap.

A.3 Unknown eccentricities

Fifteen selected targets were recomputed with the otherwise-unmeasured eccentricity Rayleigh scale set to zero or 0.15, retaining reported eccentricity measurements. All other simulation rules, the base seed and sampling counts remain unchanged. Targets with measured eccentricity need not change. The resulting reference-scenario values are in Table 9; the full scenario outputs are retained. A larger unknown eccentricity can improve angular access near an inner working angle and also alter the flux-residence gate. It is not an unqualified habitability benefit.

At scale zero, Proxima is first in 92.13% of the southern scenarios; the largest top-five frequency change is 0.93 percentage points. The corresponding portfolio is GJ 887 d, Proxima Cen b, Wolf 1061 c. At scale 0.15, Proxima is first in 92.13% of the southern scenarios; the largest top-five frequency change is 2.31 percentage points. The corresponding portfolio is GJ 251 c, GJ 887 d, Proxima Cen b.

A.4 An eccentric longer-horizon target

For HD 20794 f at its nominal eccentricity 0.45, instantaneous flux ranges from 0.178 to 1.238 Earth units, while the orbital mean is 0.419. Only 32.2% of orbital time lies in the adopted conservative

Table 9: Sensitivity of the baseline conditional score to the default Rayleigh eccentricity scale, for selected targets. Reported eccentricity measurements override this default.

Planet	Scale 0	Scale 0.05	Scale 0.15
Proxima Cen b	0.408	0.408	0.4
Gl 725 B c	1.32×10^{-3}	1.41×10^{-3}	2.21×10^{-3}
GJ 887 d	1.70×10^{-4}	1.83×10^{-4}	3.48×10^{-4}
GJ 1061 d	2.69×10^{-5}	2.69×10^{-5}	2.69×10^{-5}
GJ 1002 c	1.30×10^{-6}	1.21×10^{-6}	9.67×10^{-7}
Teegarden's Star c	1.66×10^{-7}	1.66×10^{-7}	1.66×10^{-7}
GJ 251 c	4.39×10^{-6}	4.51×10^{-6}	5.89×10^{-6}

flux interval at the nominal stellar temperature. This numerical residence diagnostic should not be interpreted as the fraction of an orbit during which life exists: heat capacity, greenhouse response and atmospheric circulation are absent. It explains why orbit-mean and persistence criteria can disagree even for an attractive angularly resolved target.

B Parameter rules and supplementary diagnostics

The polynomial used for a habitable-zone boundary is $S_{\rm eff} = S_0 + aT' + bT'^2 + cT'^3 + dT'^4$, where $T' = T_* - 5780$ K after clamping to 2,600–7,200 K. Table 10 reproduces the one-Earth-mass coefficients used in code (Kopparapu et al., 2014). A repeated boundary calculation at a different assumed planet mass would be a different scenario, not a correction to these explicitly specified ones.

Table 10: Habitable-zone coefficients used for all planets as environmental templates.

Boundary	S_0	a	b	c	d
Recent Venus	1.776	2.136×10^{-4}	2.533×10^{-8}	-1.332×10^{-11}	-3.097×10^{-15}
Runaway	1.107	1.332×10^{-4}	1.580×10^{-8}	-8.308×10^{-12}	-1.931×10^{-15}
Maximum GH	0.356	6.171×10^{-5}	1.698×10^{-9}	-3.198×10^{-12}	-5.575×10^{-16}
Early Mars	0.320	5.547×10^{-5}	1.526×10^{-9}	-2.874×10^{-12}	-5.011×10^{-16}

For a self-normalized importance estimate, $\widehat{A} = \sum \omega_i w_i / \sum \omega_i$, $\widehat{B} = \sum \omega_i D_i / \sum \omega_i$, and $\widehat{J} = \sum \omega_i w_i D_i / \sum \omega_i$. The paired covariance estimate is $\widehat{\Delta} = \widehat{J} - \widehat{A}\widehat{B}$. We use the physical-draw influence term $(w_i - \widehat{A})(D_i - \widehat{B}) - \widehat{\Delta}$, weighted by ω_i, for its Monte Carlo standard error. The joint estimator uses $w_i D_i - \widehat{J}$. These quantify numerical integration error under the declared model, not uncertainty over the model itself. The finite weights in Eq. (7) avoid the unbounded variance that a naive $1/\sin i$ reweighting of purely isotropic draws would produce for the $\gamma = 0$ case.

For inclination density proportional to $\sin^\gamma i$ over $[0, \pi/2]$, normalization integrals are $(\pi/2, 1, \pi/4)$ and $\mathbb{E}[\cos i] = (2/\pi, 1/2, 4/(3\pi))$ for $\gamma = (0, 1, 2)$. The independent test uses 400,000 mixture draws. The eccentric-flux check uses 100,000 deterministic mean-anomaly nodes. The spectral test projects out a constant and log-wavelength slope, whitens the remaining covariance, and works in the span of template differences. Discarded orthogonal noise cancels exactly in likelihood ratios; the low-dimensional implementation is therefore a sufficient-statistic speedup rather than an approximation to the specified Gaussian test.

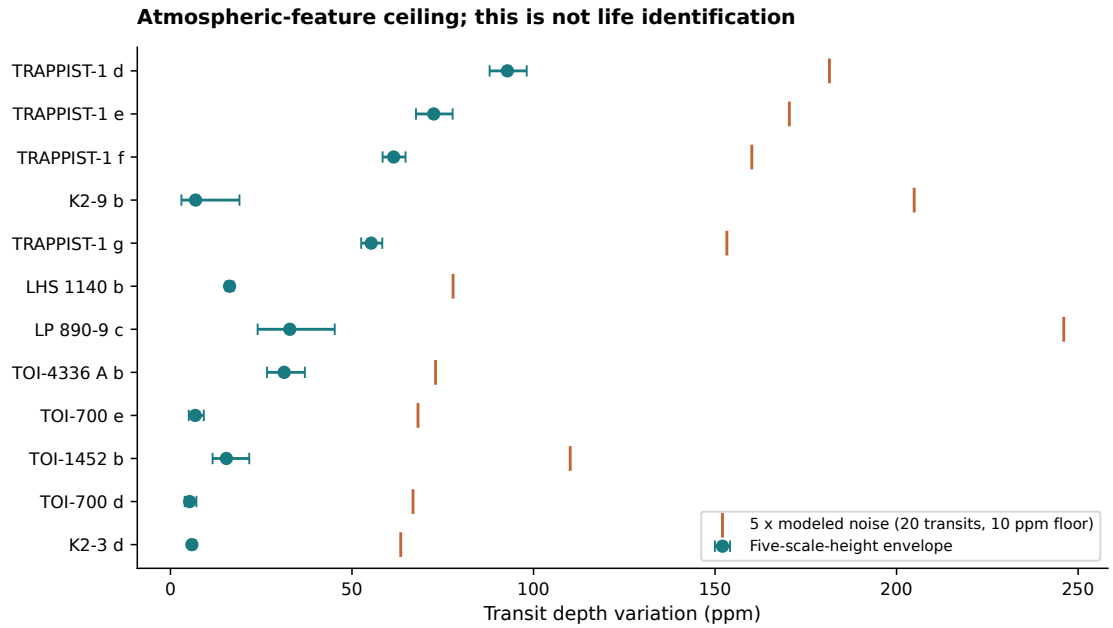


Figure 8: Selected transmission envelopes under the standard-size, empirical-zone scenario, with no feature attenuation and a 10 ppm residual floor. Markers show median five-scale-height amplitudes and 16th–84th percentile spread; vertical strokes mark five times the modeled 20-transit noise. Sorting is by the conditional feature score, not measured habitability. Unknown masses and high-mean-molecular-weight assumptions limit the interpretation.

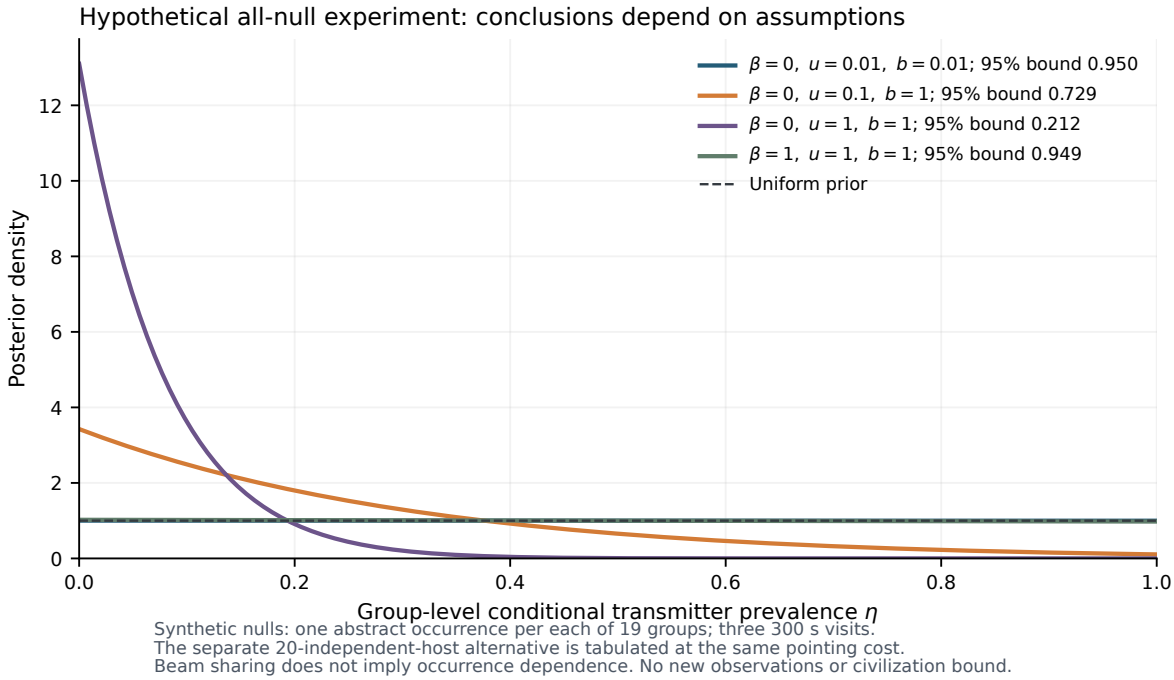


Figure 9: Posterior examples after explicitly hypothetical all-null radio observations. The displayed group-occurrence model has nineteen abstract latent groups; the numerical supplement also provides twenty independent-host trials at the same nineteen-pointing cost. A shared beam is not evidence for occurrence dependence. Distributions close to uniform illustrate a nearly uninformative null for the specified weak-emitter or low-duty population.

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